NOP → No operation

HLT

WAP in the assembly of 8085 to arrange given set of data in descending order.

WAP to sort given set of nos.

JMP Label

Counters & Delay

Clock freq of the system = $f = 2MHz$

$$T = \frac{1}{f} = \frac{1}{2 \times 10^6} = 0.5\,\mu s$$

```
    MVI C, FFH  → 7T
Loop: DCR C     → 4T
    JNZ Loop    → 10T / 7T
```

$14T \times 255 = 14 \times 255 \times 0.5$

$= 1785\,\mu s$

$3T = 3 \times 0.5$

$= 1.5\,\mu s$

7T

Total $= 1785 - 1.5 + 2 \times 0.5$

$= 1787\,\mu s$

B or C → 0000
0001

T_0 LXI B, 2384H 10T 2384 → (9092)$_{10}$

Loop: DCX B 6T

T_L MOV A, C 4T

ORA B 4T

JNZ Loop 10T/7T

Loop = 6+4+4+10 = 24T

$T_L = 24 \times 0.5 \times 9092 - 1.5$

$= 109102.5$

$T = T_L + T_0$

$= 109102.5 + 10 \times 0.5$

$= 109107.5\ \mu s \simeq 109\ ms$

8 MVI B, 38H → 56 7T

Loop2: MVI C, FFH

Loop1: DCR C → 1783.5

JNZ Loop1

DCR B → 4T

JNZ Loop2 → 10T

$T_L = (1787 + 2) \times 56$

$= 1789 \times 56$

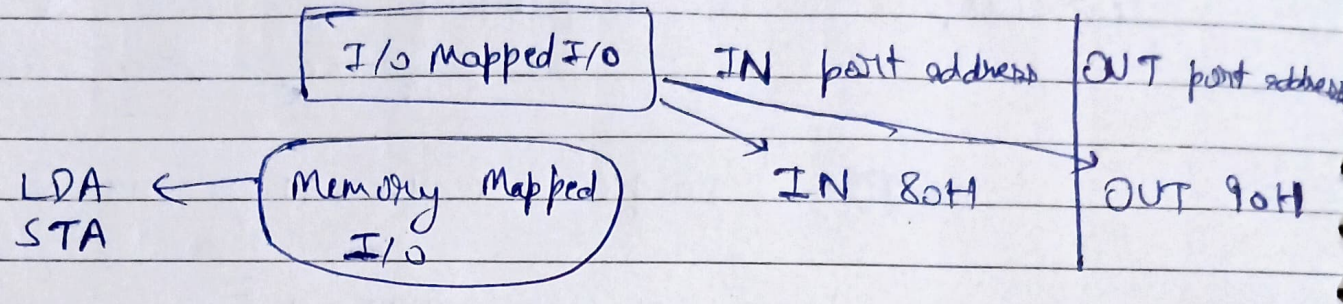
$(1789 + 10) \times 56 - 8 \times 5 =$

$T = (1783.5 + 21 \times 0.5) \times 56 - 1.5\ MS + 7.5\ \mu s$

$= ((1783.5 + 10.5) 56 + 2)\ \mu s$

$= 100,466\ \mu s \simeq 100.5\ ms$

Q WAP to count continuously in hexadec. from FFH to 00H in a system with 0.5 μs clock period. Use register C to set up 1ms delay b/w each count & display the no. at one of the output port.



Ans

```
        MVI B, 00H
NEXT:   DCR B       - 4T
        MVI C, count - 7T
delay:  DCR C       }
        JNZ delay   } TL
        MOV A, B    - 4T
        OUT port add. - 10T
        JMP Next    - 10T
```

$35T + count \times 14T = 1\,ms$

$35 \times 0.5 + 14 \times 0.5 \times count = 1000$

$$count = \frac{1000 - 35 \times 0.5}{7}$$

$$count \approx (140)_{10} = (8C)_{16}$$

Assignment

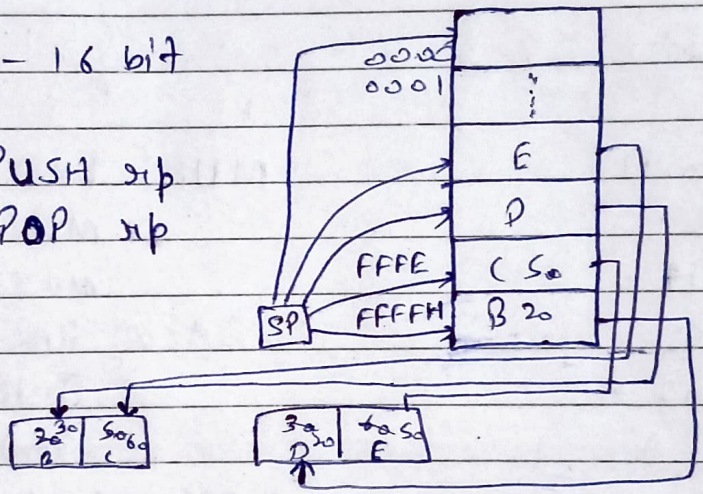
Q WAP to count from 0 to 9 with 1 sec. delay b/w each count. (clock freq. = 1 MHz)

Q WAP to generate a continuous square wave with a period of 500 μs. System clock period is 325 ns. Use bit D_0 to output the square wave.

Stack & subroutine

SP - 16 bit

PUSH rp
POP rp




```
LXI D, 3060H
LXI SP, 2000H
LXI B, 2050H

PUSH B
PUSH D
POP B
POP D
```

PUSH PSW → It tells the status of flags
POP PSW

PSW

A	F

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
S	Z	x	Ac	x	P	x	CS

Not used

WAP to perform the functions:

Ans (i) Clear all flags (ii) load 00H in the accumulator & demonstrate zero flag is not affected by data transfer instructions (iii) Logically OR the acc. with itself to set the zero flag & display the flag at Port 1 or store all the flags on the stack.

Subroutine

For multiplication

```
LHLD 2100H
MOV  E,L
MOV  A,H
CALL MUL
SHLD 2102H
HLT
```

```
MUL:  LXI H,0000H
      MVI D,00H
      MVI C,08H
BACK: RAR
      JNC NEXT
      DAD D
NEXT: XCHG
      DAD H
      XCHG
      DCR C
      JNZ BACK
      RET
```

Conditional call

CC : Call on carry
CNC : Call on No carry

Similarly, CZ, CNZ, CM, CP, CPE, CPO
(Zero/Non-zero, -ve/+ve, Parity)

Conditional Return

RC, RNC, RZ, RNZ, RM, RP, RPE, RPO
(carry flag, zero flag, sign flag, Parity)

PCHL → PC ← HL

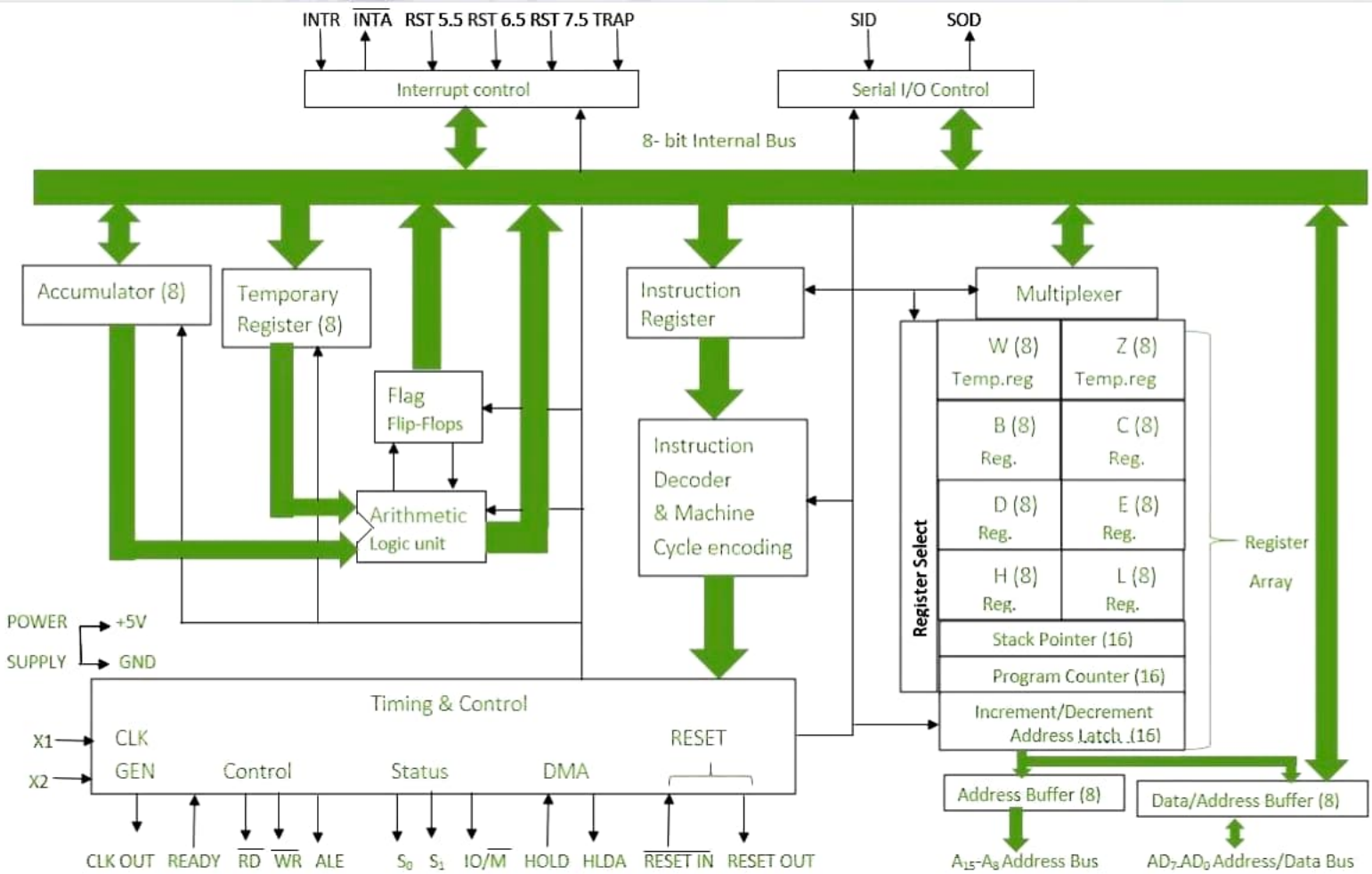
XTHL (EXchange top of stack with HL rp) (Z)

SPHL → SP ← HL

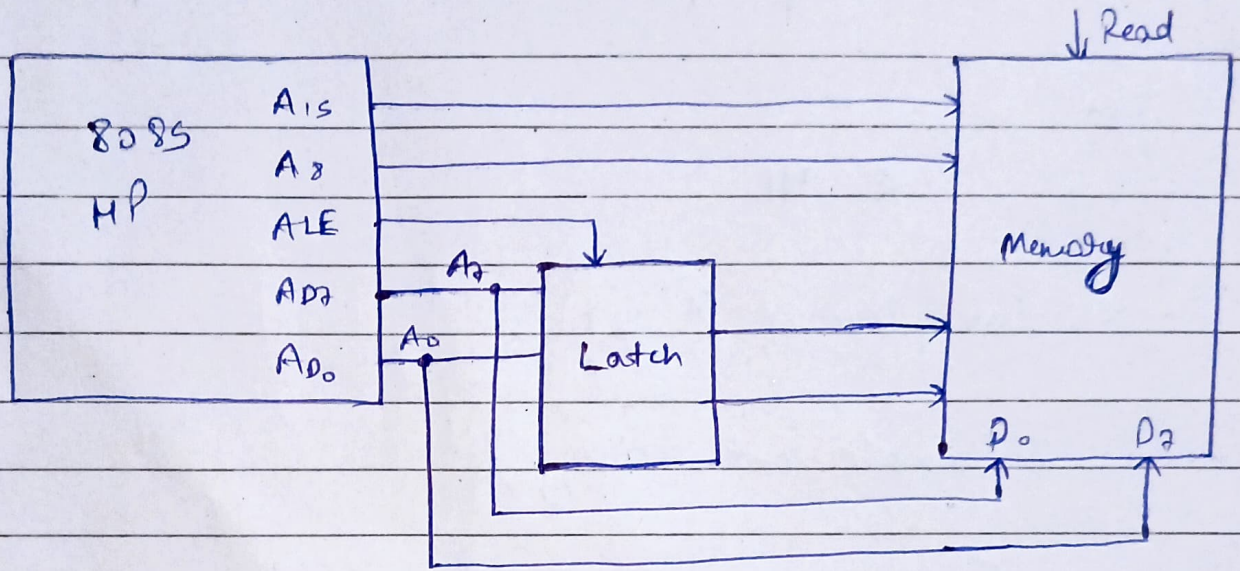
CMA → (Complement Acc.)

CMC → (complement carry)

STC → Set carry


INTR
INTA
RST 5.5 RST 6.5 RST 7.5 TRAP
SID
SOD
Interrupt control
Serial I/O Control
8- bit Internal Bus
Accumulator (8)
Temporary Register (8)
Flag Flip-Flops
Arithmetic Logic unit
Instruction Register
Instruction Decoder & Machine Cycle encoding
Multiplexer
W (8) Temp.reg
Z (8) Temp.reg
B (8) Reg.
C (8) Reg.
D (8) Reg.
E (8) Reg.
H (8) Reg.
L (8) Reg.
Register Select
Register Array
Stack Pointer (16)
Program Counter (16)
Increment/Decrement Address Latch (16)
POWER SUPPLY
+5V
GND
Timing & Control
X1
X2
CLK GEN
Control
Status
DMA
RESET
Address Buffer (8)
Data/Address Buffer (8)
CLK OUT
READY
RD
WR
ALE
S0
S1
IO/M
HOLD
HLDA
RESET IN
RESET OUT
A15-A8 Address Bus
AD7-AD0 Address/Data Bus

Address Latch Enable



- First machine cycle → carry address
- After one cycle, $AD_7 - AD_0$ → storing data
- All this possible by ALE

ALE → storing address somehow

Pins (40 pin)

One pin for voltage

HLDA → Hold acknowledge } for supporting DMA
HOLD

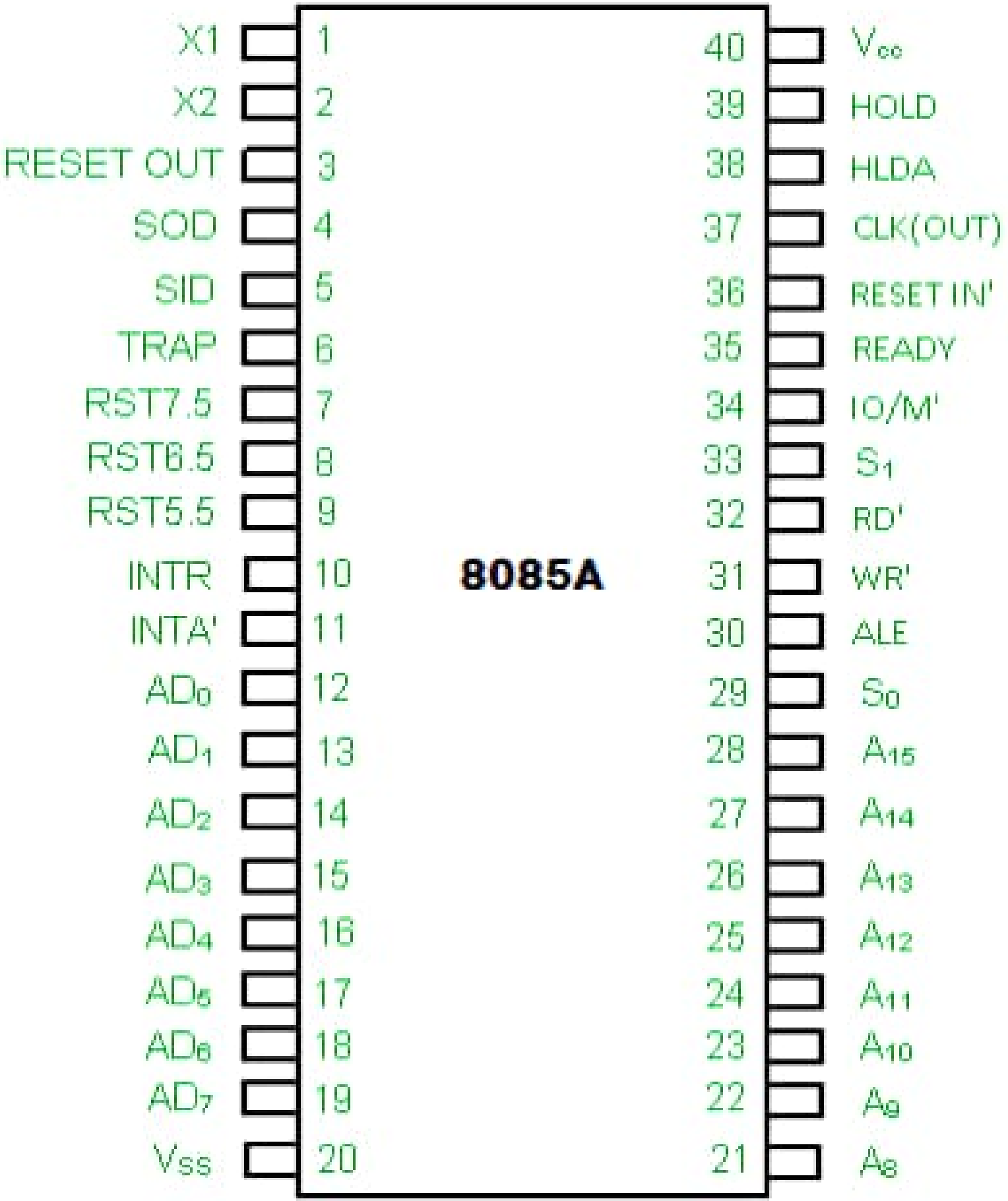
S_0, S_1 ⟶ to check if it is instruction fetch or operand fetch

Status signals

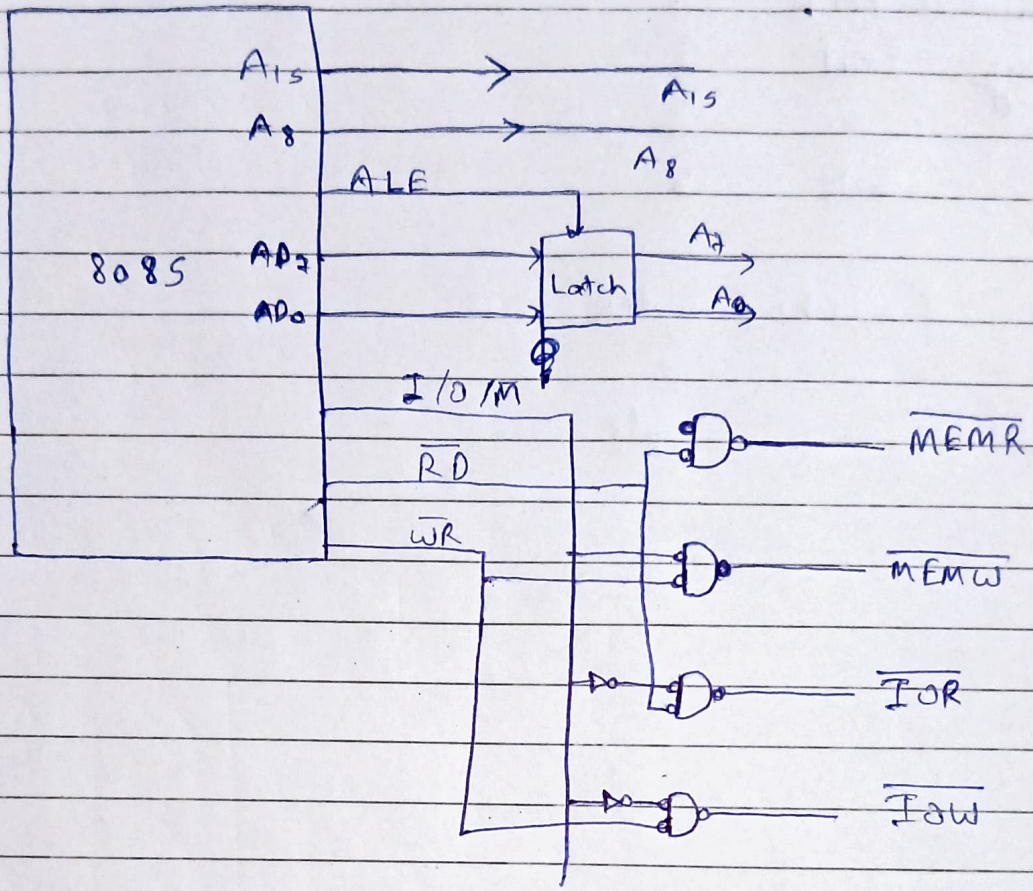
M/C cycle	IO/M	S_1	S_0	control Signal
opcode fetch	0	1	1	$\overline{RD} = 0$
Memory read	0	1	0	$\overline{RD} = 0$
" write	0	0	1	$\overline{WR} = 0$
I/O read	1	1	0	$\overline{RD} = 0$
" write	1	0	1	$\overline{WR} = 0$
Interrupt acknowledge	1	1	1	INTA = 0

- Read & write are high impedence state

Reset	Z	x	x
Halt	Z	0	0
Hold	Z	x	x


X1 1
X2 2
RESET OUT 3
SOD 4
SID 5
TRAP 6
RST7.5 7
RST6.5 8
RST5.5 9
INTR 10
INTA' 11
AD0 12
AD1 13
AD2 14
AD3 15
AD4 16
AD5 17
AD6 18
AD7 19
VSS 20
8085A
40 VCC
39 HOLD
38 HLDA
37 CLK(OUT)
36 RESET IN'
35 READY
34 IO/M'
33 S1
32 RD'
31 WR'
30 ALE
29 S0
28 A15
27 A14
26 A13
25 A12
24 A11
23 A10
22 A9
21 A8

Instruction Cycle = Instruction Fetch + Execute cycle



Machine Cycle

Time reqd to complete one operation of memory access, I/O, acknowledge external request.

LDA 2050 H

1. Instruction Fetch 4
2. Memory read 3
3. " " 3
4. " " 3 (Accumulator)

STA 2060 H

1. Instruction Fetch 4
2. Memory Read 3
3. " " 3
4. " write 3

IN - 3 machine cycles
OUT - " " "
MOV A,B - 1 " cycle